

Experiment 8

Nmap Service Version and OS Detection

Aim

To perform service version detection and operating system identification using Nmap.

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands

i) Detect Service Versions

```
nmap -sV 192.168.1.10
```

Description:

Determines the versions of services running on target ports.

ii) Aggressive Scan

```
nmap -A 192.168.1.10
```

Description:

Performs OS detection, version detection, script scanning, and traceroute.

iii) Operating System Detection

```
sudo nmap -O 192.168.1.10
```

Description:

Identifies the operating system running on the target host.

Sample Output

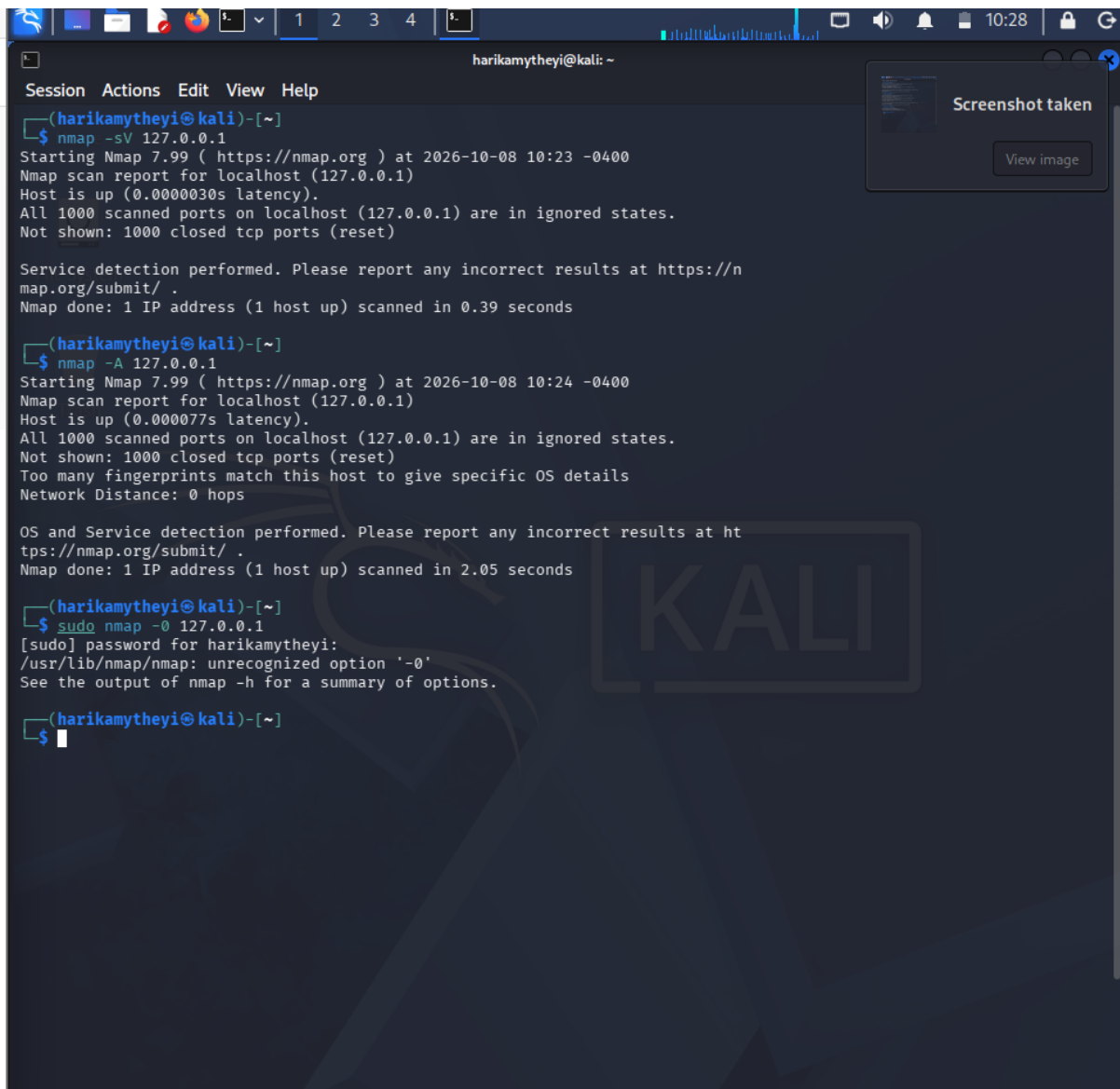
```
PORT      STATE SERVICE VERSION
```

```
22/tcp    open  ssh      OpenSSH 8.2
```

```
80/tcp    open  http     Apache 2.4
```

Result

Successfully performed service version detection and OS identification using Nmap.



```
(harikamytheyi@kali)-[~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-08 10:23 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds

(harikamytheyi@kali)-[~]
$ nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-08 10:24 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000077s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.05 seconds

(harikamytheyi@kali)-[~]
$ sudo nmap -0 127.0.0.1
[sudo] password for harikamytheyi:
/usr/lib/nmap/nmap: unrecognized option '-0'
See the output of nmap -h for a summary of options.

(harikamytheyi@kali)-[~]
$
```